

K-LC6 Doppler Signal Conditioning

Filter and Amplifier Design Rationale

Non-contact Ground Speed Sensor (NGSS) — Formula Student

1. Overview

The K-LC6-RFB-00D 24 GHz Doppler radar module produces I and Q baseband signals at its IF outputs ranging from sub-microvolt to a few millivolts in amplitude, sitting on a typical 0.2 V DC offset. These signals are far too small to drive an analog-to-digital converter (ADC) directly, and they sit in a wide bandwidth (0–10 MHz at the sensor) that contains far more noise than useful Doppler signal. A dedicated signal conditioning chain is therefore required between the sensor and the FPGA's ADC.

The conditioning chain implemented here is a two-stage active band-pass amplifier, based on the topology in RFbeam Application Note AN-04 but with component values adapted for the ground speed sensing application (stationary to ~100 km/h vehicle speed). The chain is built per channel: one instance for the I output, one for the Q output, replicated for each of the four radar modules in the eventual Janus configuration (eight channels total in the production build).

1.1 Block-level signal chain

The chain comprises four conceptual elements, in order:

- Input AC coupling — high-pass filter that blocks the sensor's DC offset and rejects sub-Hz drift.
- Stage 1 inverting amplifier — gain of 20 dB with integrated low-pass roll-off in feedback.
- Inter-stage AC coupling — high-pass filter that blocks the amplified DC and drift from Stage 1.
- Stage 2 inverting amplifier — additional gain (20–26 dB depending on build) with low-pass roll-off.

All op-amps operate from a single +5 V rail with their non-inverting inputs biased to 2.5 V (mid-rail), allowing the AC signal to swing symmetrically around 2.5 V without clipping.

2. Why Filtering Is Required

2.1 Removing the sensor DC offset

The K-LC6 datasheet specifies a typical IF output DC offset of 0.2 V on both I and Q channels. This offset is not signal — it is the quiescent operating point of the mixer's baseband output when no target return is present. If this DC level were amplified by the full chain gain (40–47 dB, factor of 100–224), it would produce an output DC level of 20–45 V, far exceeding the +5 V supply rail. Even op-amp input offset voltages of a few millivolts, multiplied by chain gain, would saturate the output stage.

The input AC coupling capacitor (high-pass filter formed with the first-stage input resistor) blocks all DC. Only the AC content — the actual Doppler signal — passes through to be amplified. The inter-stage capacitor performs the same role between Stage 1 and Stage 2: it blocks the (small but amplified) DC offset and bias-current-induced offset at Stage 1's output from being re-amplified by Stage 2.

2.2 Suppressing 1/f (flicker) noise

Every semiconductor device exhibits 1/f noise — a noise power spectral density that rises as frequency decreases. Above approximately 10 Hz, the noise floor is dominated by flat thermal (white) noise. Below 10 Hz, the 1/f component climbs rapidly. By placing the high-pass filter corner at a few hertz, the worst of the 1/f noise region is excluded from the passband. This improves the signal-to-noise ratio for the actual Doppler frequencies of interest, which all sit well above the corner.

2.3 Rejecting mechanical and vibrational disturbances

When the sensor is mounted on a moving vehicle, the sensor itself undergoes accelerations, vibrations, and suspension-induced motions. Vehicle pitch, ride-height changes, and chassis flex all produce slow Doppler-like signatures from stationary clutter (the ground, nearby walls, the wheel arch) as the sensor's geometry relative to those objects changes. The high-pass filter at 2–5 Hz attenuates these mechanically-induced signatures while leaving the genuine ground speed signal (which begins above 220 Hz at 5 km/h) untouched.

2.4 Limiting bandwidth to the signal of interest

The K-LC6 IF output has a native bandwidth of 0–10 MHz (–3 dB). Almost all of this bandwidth contains only thermal noise — there is no Doppler content above the maximum target speed. Signal-to-noise ratio scales inversely with bandwidth: every doubling of bandwidth admits 3 dB more noise without admitting any more signal. AN-04 explicitly notes that limiting bandwidth to the real needs is essential for detection range. The low-pass filters in each stage roll the response off above the maximum Doppler frequency expected, dramatically improving SNR for the FFT-based spectral peak detection that follows in the FPGA.

2.5 Anti-aliasing for the ADC

The low-pass filter also serves as an anti-aliasing filter for the FPGA's ADC. Any signal content above half the sampling frequency (the Nyquist limit) folds back into the baseband and corrupts the FFT result. With an LP corner at 10 kHz and a planned ADC sample rate of 30 kHz or higher, the LP filter provides adequate roll-off (–6 dB/octave, second-order with the LP corners stacked across both stages) to suppress aliases before they reach the converter.

3. Justification of Cutoff Frequencies

3.1 Doppler frequency budget

The K-LC6 produces a Doppler frequency of 44.7 Hz per km/h for head-on motion (158 Hz per m/s), derived from the standard Doppler relation $f_d = 2v \cdot \cos(\theta) / \lambda$ at 24.125 GHz. For the Janus configuration the actual signal frequency depends on the mounting angle θ :

Doppler frequencies across the operating range

Vehicle speed	Head-on ($\theta = 0^\circ$)	Janus 30°	Janus 45°
Stationary	0 Hz	0 Hz	0 Hz
5 km/h (low-speed creep)	224 Hz	194 Hz	158 Hz
30 km/h	1.34 kHz	1.16 kHz	948 Hz
60 km/h	2.68 kHz	2.32 kHz	1.90 kHz
100 km/h	4.47 kHz	3.87 kHz	3.16 kHz
120 km/h (motor rig)	5.36 kHz	4.65 kHz	3.79 kHz

The signal of interest therefore occupies the band from approximately 200 Hz (lowest meaningful speed, 5 km/h on Janus 45°) up to approximately 5.5 kHz (worst-case 120 km/h, head-on).

3.2 High-pass cutoff: ~3 Hz

Choice and reasoning

Both high-pass corners (input coupling and inter-stage coupling) are placed at approximately 3 Hz. This sits well below the lowest signal frequency of interest (~200 Hz) and provides almost two decades of margin, ensuring negligible attenuation of low-speed signals while still aggressively rolling off the $1/f$ noise region, DC offsets, and slow mechanical drift.

Why not lower?

A lower corner (e.g. 0.1 Hz) would require either much larger capacitors (impractical with available components) or much larger resistors (which raise offset-current contributions and thermal noise). It would also slow the recovery time after a saturation event — with a 3 Hz corner the chain recovers from a transient overload in roughly 50 ms, whereas a 0.1 Hz corner would take 1.5 seconds. Fast recovery is important when the sensor encounters strong specular returns or transient interference, since the FFT pipeline becomes unusable while the chain is settling.

Why not higher?

A higher corner (e.g. 50 Hz) would meaningfully attenuate the lowest creep speeds. At 5 km/h on a 45° Janus mounting, the signal frequency is 158 Hz; a 50 Hz first-order filter would attenuate this signal by approximately 1 dB and shift its phase, complicating the FFT peak detection. The 3 Hz corner ensures the entire operational speed range sees flat passband response.

3.3 Low-pass cutoff: ~10 kHz

Choice and reasoning

Both low-pass corners are placed at approximately 10 kHz. This sits comfortably above the highest expected signal frequency (~5.5 kHz at 120 km/h on the motor rig, head-on geometry) with sufficient margin for the Doppler spectral spread caused by finite antenna beamwidth, and for any future overspeed testing.

Doppler spectral spread

A real-world ground speed signal is not a single frequency line; it is a spread of frequencies produced by the antenna beam's finite width. The K-LC6's 12° beamwidth in the travel direction produces a fractional Doppler spread of approximately $0.21 \cdot \sin(\theta_i)$ of the centre frequency. For a 45° Janus mounting at 100 km/h (centre ~3.16 kHz), this yields a spread of ~470 Hz on either side of the peak. The LP corner must accommodate this skirt without attenuating it, which 10 kHz comfortably does.

Why not lower?

Setting the LP corner closer to the maximum signal frequency would clip the upper Doppler skirt and distort the spectral peak shape. The XCA (cross-correlation) and spectral centroid algorithms used for SoG estimation rely on the full spectral shape — particularly the upper edge — to locate the true mean Doppler frequency. Premature roll-off would bias the speed estimate downward.

Why not higher?

Above ~15 kHz, additional bandwidth admits more thermal noise without any signal benefit, degrading SNR. It also relaxes the anti-aliasing requirements at the ADC, requiring either a higher ADC sample rate or accepting some aliasing risk.

4. Justification of Gain Values

4.1 Required gain — first-principles calculation

The K-LC6 IF output amplitude depends on target reflectivity, distance, and geometry. For the ground speed sensing application (sensor mounted 0.3–0.5 m above a dry asphalt or concrete surface at an oblique angle), the expected AC signal at the IF pin is approximately 0.5–2 mV RMS, with worst-case weak returns dropping to ~50–100 µV RMS. The sensor noise floor in the processing bandwidth (~5 kHz) is approximately 3 µV RMS.

The downstream ADC requires signal levels of approximately 1 V peak-to-peak (~350 mV RMS) for good quantisation SNR. Achieving this from a 1 mV typical sensor signal requires a gain of approximately 350, or about 51 dB. Allowing some margin and accounting for the planned ADC input range, the design target is 40–50 dB of total chain gain.

4.2 Two-stage split

The required gain is split across two stages rather than concentrated in a single op-amp, for several reasons:

- Gain-bandwidth product (GBW) limits — the LMV772 has a GBW of approximately 3.5 MHz. A single stage with 50 dB (gain of 316) would have a closed-loop bandwidth of only $3.5 \text{ MHz} / 316 \approx 11 \text{ kHz}$, marginal for the 10 kHz signal band. Two stages of 25 dB each (gain of ~ 18) leave bandwidth comfortably above 190 kHz per stage, well clear of the 10 kHz signal corner.
- Filter ordering — each stage contributes its own first-order high-pass and low-pass filter, giving an overall second-order band-pass response. A single stage could only provide first-order roll-off in either direction, leaving more noise and aliases in the output.
- Saturation containment — if a strong specular return saturates Stage 1, the inter-stage AC coupling capacitor prevents the saturation DC level from propagating into Stage 2, and the recovery time is shorter than for a single high-gain stage.
- Offset multiplication — input-referred offset voltages and bias currents are multiplied by the full gain. Splitting gain across stages means Stage 2's offset is multiplied only by Stage 2's gain, not the full chain gain.

4.3 Why not more gain?

Higher gain (60–80 dB, as cited in AN-04 for typical person-detection applications) is unnecessary and counterproductive for ground speed sensing. The ground surface provides a distributed, relatively strong scatterer compared to a single 1 m^2 human target at 24 m (the AN-04 reference case). With excessive gain:

- Normal returns saturate the output stage, clipping the FFT input.
- Specular reflections from smooth surfaces or wet patches saturate severely and produce long recovery transients.
- Op-amp offset and noise voltage are multiplied unnecessarily.

A chain gain of 40–47 dB is well-matched to the expected signal range and leaves headroom for the strongest realistic returns without clipping.

4.4 Why not less gain?

Lower gain (20–30 dB) would leave weak returns (gravel, wet surfaces with low backscatter, long standoff) below the ADC's effective resolution. Even a 12-bit ADC with 1 V range has a quantisation step of $244 \mu\text{V}$; a $100 \mu\text{V}$ signal would be lost in the bottom bit. The 40 dB minimum ensures the weakest expected signals are still clearly resolvable above quantisation noise.

5. As-Built Component Values

The current bench build uses components on hand: $1 \mu\text{F}$ ceramic capacitors (largest value available for AC coupling) and standard resistor values. Time constants are preserved by scaling resistance to compensate for the reduced capacitance.

5.1 Per-channel component list

Ref	Value	Function	Resulting parameter
C1	1 μ F ceramic	Input HP cap	With R1 (47 k Ω): f_HP = 3.4 Hz
R1	47 k Ω	Stage 1 input	Sets Stage 1 gain (with R2) and HP corner
R2	470 k Ω	Stage 1 feedback	Gain = R2/R1 = 10 (20 dB)
C2	33 pF	Stage 1 LP cap	With R2 (470 k Ω): f_LP = 10.3 kHz
C3	1 μ F ceramic	Inter-stage HP cap	With R3 (47 k Ω): f_HP = 3.4 Hz
R3	47 k Ω	Stage 2 input	Sets Stage 2 gain (with R4) and HP corner
R4	470 k Ω	Stage 2 feedback	Gain = R4/R3 = 10 (20 dB)
C4	33 pF	Stage 2 LP cap	With R4 (470 k Ω): f_LP = 10.3 kHz
R5	100 Ω	Output series R	Output protection and cable damping
R10, R11	470 k Ω each	Bias divider	Generates 2.5 V mid-rail reference
C10	10 μ F electrolytic	Bias bypass	Low-impedance mid-rail reference for AC

5.2 Resulting chain parameters

Parameter	Value
Total gain	40 dB ($\times 100$)
High-pass corner (each)	3.4 Hz
Low-pass corner (each)	10.3 kHz
Effective passband (-3 dB)	~ 5 Hz to ~ 8 kHz (combined two-stage)
Quiescent output level	2.5 V (mid-rail)
Output swing capability	± 2.5 V (rail-to-rail op-amp)
Expected output for typical SoG return	~ 600 mV pk-pk

5.3 Path to final production values

When better components are available for the production PCB build, the chain can be upgraded to approximately 47 dB by changing R4 to 1 M Ω (the AN-04-stated ceiling for offset/noise reasons) and C4 to 15 pF to maintain the 10 kHz LP corner. This pushes Stage 2 to 26.5 dB while keeping Stage 1 at 20 dB. The total of ~ 47 dB sits in the middle of the design target band. The 15 pF capacitor is acceptable on a controlled-stray PCB but marginal on veroboard where parasitic capacitance is on the same order, which is why the bench build sits at 40 dB.

6. Supporting Design Choices

6.1 Op-amp selection: LMV772

The LMV772 is recommended explicitly in AN-04 for the low-noise variant of this topology. Relevant characteristics:

- Single-supply rail-to-rail input/output operation, compatible with the +5 V single rail.
- Input voltage noise of 14 nV/VHz at 1 kHz, well below the sensor's own noise contribution (45 nV/VHz at the IF output).
- Gain-bandwidth product of 3.5 MHz, sufficient for 20 dB gain with a 10 kHz corner.
- Input offset voltage of typically 1 mV, manageable across the two-stage chain.
- Dual op-amp package (8-pin SOIC or DIP), one package per channel (two stages).

6.2 Mid-rail bias network

Operating on a single +5 V supply requires the op-amp's quiescent output to sit at a level permitting symmetric AC swing. The 470 k Ω / 470 k Ω resistive divider generates a 2.5 V reference tied to the non-inverting input of each stage. The 10 μ F electrolytic from this node to ground provides a low AC impedance, presenting an effective AC ground at the non-inverting input without disturbing the DC bias point. This means each stage sees its non-inverting input as a rock-solid 2.5 V for AC purposes, while DC current draw remains negligible (~ 5 μ A through the divider).

6.3 Power supply decoupling

AN-04 explicitly recommends linear regulation over switching regulation for high-gain Doppler amplifiers. Switching regulator ripple, even at modest amplitudes, appears at the amplifier output multiplied by chain gain — a 1 mV switching spur becomes 100 mV at the output and creates spurious peaks throughout the FFT bin range. The bench build uses linear regulation from the bench supply. Decoupling capacitors (100 nF ceramic at each op-amp Vcc pin, 10 μ F bulk at the sensor Vcc pin) are used to suppress any residual high-frequency noise on the rails.

6.4 Differences from AN-04 reference design

The AN-04 reference values target person detection (4 Hz – 400 Hz passband). The adapted values target ground speed sensing across a wider Doppler frequency range. Key differences:

- Low-pass corner moved from ~ 1.6 kHz to ~ 10 kHz to accommodate higher Doppler frequencies (100+ km/h target speed vs. ~ 10 km/h person speed).
- Total gain reduced from 73 dB to 40 dB because the ground surface return is much stronger than a 1 m² person at 24 m range.
- High-pass corner moved from 4 Hz to 3 Hz, marginally lower to ensure flat response across the operational speed range without unnecessarily widening the passband.

7. Summary

The signal conditioning chain is a two-stage AC-coupled band-pass amplifier operating from a single +5 V rail. It performs four functions: (1) removing the sensor's 0.2 V DC offset, (2) suppressing 1/f noise and mechanical disturbances below 3 Hz, (3) limiting bandwidth to ~10 kHz to maximise SNR and provide anti-aliasing for the ADC, and (4) amplifying the microvolt-to-millivolt Doppler signal up to a level suitable for the FPGA's ADC.

Cutoff frequencies are chosen to bracket the operational Doppler frequency range (200 Hz – 5.5 kHz) with appropriate margin: high-pass at ~3 Hz (well below the lowest signal frequency) and low-pass at ~10 kHz (well above the highest signal frequency, allowing for beamwidth-induced spectral spread). Gain is set at 40–47 dB to match the ground surface return amplitude expected at typical sensor mounting heights and the dynamic range of the downstream ADC.

The current bench build implements 40 dB of gain with a 3.4 Hz / 10.3 kHz passband, using components on hand. The architecture and topology match the production PCB build; only specific component values change. Once veroboard validation is complete, the same circuit is replicated for the Q channel of the same module, then for the I and Q channels of the remaining three modules in the Janus configuration.